Approximation Error of Polynomial Interpolants

Now we know how to construct an interpolating polynomial $p_{m-1}(t)$ of degree at most $m-1$ that interpolates a data set $\{(t_i, y_i)\}_{i=1}^{m}$ or $\{(t_i, f(t_i))\}_{i=1}^{m}$

For our first data set, we have no way of knowing how well $p_{m-1}(t)$ approximates the process that generates the data. All we know is that it duplicates the data at the given nodes. <Could hold back data for comparison>.

If our data comes from sampling a complicated function $f(t)$, though, it turns out that we can derive an expression for

$$\| f(t) - p_m(t) \|_\infty = \max_{t \in [a,b]} | f(t) - p_{m-1}(t) |$$

Before we get to the result, here is some terminology.

<u>def$^{\underline{n}}$</u> The spread of a set of points $\{s_1, s_2, \ldots, s_m\}$, denoted $\text{spr}\{s_1, \ldots, s_m\}$ is the smallest interval containing $\{s_1, \ldots, s_m\}$

The open spread of a set of points $\{s_1, s_2, \ldots, s_m\}$, denoted $\text{ospr}\{s_1, \ldots, s_m\}$, is the largest open interval that is a subset of $\text{spr}\{s_1, \ldots, s_m\}$.

note: if points ordered with $s_1 \leq s_2 \leq \cdots \leq s_m$ then $\text{spr}\{s_1, s_2, \ldots, s_m\} = [s_1, s_m]$ and $\text{ospr}\{s_1, s_2, \ldots, s_m\} = (s_1, s_m)$

Main result:

<u>Theorem</u> (error in polynomial interpolation):
If $f(t)$ has m continuous derivatives and $p_{m-1}(t)$ is the polynomial of degree at most $m-1$ that interpolates f at the distinct points $t_1, t_2, \ldots, t_m$, then, for any t,

$$f(t) - p_{m-1}(t) = \frac{f^{(m)}(c_{\hat{t}})}{m!} \prod_{j=1}^{m} (t - t_j)$$

minus sign of Taylor error

where $c_{\hat{t}}$ is an unknown point in $\text{ospt}\{t_1, \ldots, t_m, t\}$ that depends on t.

<u>Proof</u>: Let $\hat{t} \neq t_i$ be some fixed point and consider the function

$$g(t) = \underbrace{f(t)}_{\text{original function}} - \underbrace{p_{m-1}(t)}_{\text{our interpolant}} - \underbrace{\left[f(\hat{t}) - p_{m-1}(\hat{t})\right]}_{\text{constant for given } \hat{t}} \underbrace{\prod_{j=1}^{m} \frac{(t - t_j)}{(\hat{t} - t_j)}}_{\text{a poly of degree at most } m}$$

Let's evaluate $g(t)$ at $t = t_i$, $i = 1:m$.

$$g(t_i) = \underbrace{f(t_i) - p_{m-1}(t_i)}_{0} - \left[f(\hat{t}) - p_{m-1}(\hat{t})\right] \prod_{j=1}^{m} \frac{(t_i - t_j)}{(\hat{t} - t_j)}$$

$$= 0 - [\quad] \cdot 0$$

$= 0$

1 term in num. is 0

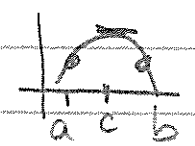
$\therefore$ $g(t)$ has roots at $t = t_i$, $i = 1:m$.

Also, $g(\hat{t}) = f(\hat{t}) - p_{m-1}(\hat{t}) - \left[f(\hat{t}) - p_{m-1}(\hat{t})\right] \cdot \left(\prod_{j=1}^{m} \frac{(\hat{t}-t_j)}{(\hat{t}-t_j)}\right)$

$$= [\quad] - [\quad] \cdot 1$$
$$= 0.$$

$\therefore$ $g(t)$ also has a root at $t = \hat{t}$.
In total, $g(t)$ has at least $m+1$ roots.

Aside:
We have seen Rolle's Theorem:
If $h \in C^1[a,b]$ and $h(a) = h(b)$ then $\exists c \in (a,b)$ such that $h'(c) = 0$.



Rolle's theorem can be generalized:
$h(a) = h(b) = h(c)$ then $\exists\, d \in \text{ospr}\{a,b,c\}$ such that $h''(d) = 0$

In general, $h(t_i) = h(t_j)$ for $i, j = 1:n$ then $\exists\, c \in \text{ospr}\{t_i\}_{i=1}^{n}$ s.t.

$$h^{(n-1)}(c) = 0.$$

back to our problem $g(t)$ has $m+1$ roots.

$\therefore$ $\exists$ a point $c_{\hat{t}} \in \text{ospr}\{t_1, \ldots, t_m, \hat{t}\}$ such that

$$g^{(m)}(c_{\hat{t}}) = 0.$$

We have $0 = g^{(m)}(c_{\hat{t}})$

Since P_{m-1} is poly of degree at most m

$$= f^{(m)}(c_{\hat{t}}) - 0 - \left[f(\hat{t}) - P_{m-1}(\hat{t})\right] \frac{d^m}{dt^m}\left[\prod_{j=1}^{m} \frac{t-t_j}{\hat{t}-t_j}\right]\Bigg|_{@t=c_{\hat{t}}}$$

Now $\displaystyle\prod_{j=1}^{m}\left(\frac{t-t_j}{\hat{t}-t_j}\right) = \underbrace{\frac{1}{\prod_{j=1}^{m}(\hat{t}-t_j)}}_{\text{a constant}} \underbrace{\prod_{j=1}^{m}(t-t_j)}_{\text{a monic polynomial of degree } m}$

$c_{m-1} = 1$

$$\therefore \frac{d^m}{dt^m}\left(\;\right) = \frac{1}{\prod_{j=1}^{m}(\hat{t}-t_j)} \cdot (m!).$$

$$\therefore \quad f(\hat{t}) - P_{m-1}(\hat{t}) = \frac{f^{(m)}(c_{\hat{t}})}{m!} \prod_{j=1}^{m}(\hat{t}-t_j)$$

This holds for any $\hat{t} \neq t_i$. It is clearly also true for $\hat{t} = t_i$ $(0 = 0)$.

We have, then, for an interval $[a,b]$ that contains $\operatorname{spr}\{t_i\}_{i=1}^{m}$ then

$$\max_{t\in[a,b]} \left| f(t) - P_{m-1}(t) \right| \leq \frac{1}{m!} \cdot \max_{t\in[a,b]} \left| f^{(m)}(t) \right| \cdot \max_{t\in[a,b]} \left| w_m(t) \right|$$

where $w_m(t) = \prod_{j=1}^{m} (t - t_j)$.

note: the error depends on

- the number of interpolation points m via $\left(\frac{1}{m!}\right)$
- the size of the mth derivative
- the placement of the nodes $\{t_j\}_{j=1}^{m}$

- we can use this result to determine bounds on the approximation error.